NAZMUL HUDA BADHON

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EDUCATION

Bachelor of Science	Computer Science and Engineering Daffodil International University, CGPA: 3.88 / 4.00	Spring 2022–Fall 2025 Dhaka, Bangladesh
	Exchange Semester (Computer Engineering) Dongseo University, GPA: 3.94 / 4.50	Fall 2023 Busan, South Korea

RESEARCH INTERESTS

Deep Learning, Convolutional Neural Networks, Medical AI.

SCHOLARSHIPS AND AWARDS

Global Korea Scholarship (GKS) for Exchange Study | Government of the Republic of Korea	Fall 2023
Special Scholarship for Consistent Academic Excellence | Daffodil International University	Spring 2022–Fall 2025
Research Award for Outstanding Undergraduate Research Contribution | Division of Research, DIU	2024
Peer Reviewer Recognition (Digital Health) | SAGE Publishing	2025

RESEARCH PUBLICATIONS

RESEARCH ARTICLES

1. **Nazmul Huda Badhon**, Imrus Salehin, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction. Global Energy Interconnection, Elsevier, 14 Nov 2025. ISSN: 2590-0358.
2. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. MethodsX, 103680, Elsevier, 17 Oct 2025. ISSN: 2215-0161.

REVIEW ARTICLES

3. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. Engineering Reports 7.9: e70365, Wiley, 12 Sep 2025. ISSN: 2577-8196.

DATA ARTICLES

4. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. Data in Brief, 110083, Elsevier, 23 Jan 2024. ISSN: 2352-3409.

BOOK CHAPTER

5. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. Engineering Applications of AI for Demand Forecasting, Taylor & Francis. Accepted on 24 Apr 2025 (In press).

PROJECTS

SELECTED RESEARCH PROJECTS

Breast Cancer Classification using Hybrid Segmentation Deep Learning

- Built a two-stage U-Net and CNN-based hybrid model on the BUSI datasets.
- Achieved strong performance on accuracy and ROC metrics.

Lung and Colon Cancer Histopathology Classification

- Applied preprocessing techniques on histopathology images and evaluated multiple transfer learning models.
- Identified the best-performing model by accuracy and F1-score.

Vision Transformer–Based Medical Image Classification

- Designed preprocessing workflows for medical images and performed classification with ViT.

- Achieved 87% testing accuracy and improved transformer performance.

ADDITIONAL PROJECTS

Lost and Found Web Application

- Developed a full-stack web app with HTML, CSS, and JS for reporting and searching lost and found items.
- Implemented PHP-based backend, REST APIs, MySQL database, and secure user/admin authentication with image upload support.

TECHNICAL SKILLS

- **Programming:** Python, Java, C
- **Machine Learning & Deep Learning:** Convolutional Neural Networks, U-Net, Vision Transformers, Transfer Learning
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Tools & Platforms:** GitHub, Google Colab, LaTeX

EXTRA-CURRICULAR ACTIVITIES

- Team Leader, International Camp | Dongseo University, South Korea — 2023
  Led and coordinated a multicultural team of 20 international students during academic and cultural programs.
- Volunteer Internship, International Affairs | Daffodil International University — 2024
  Assisted in international student support, coordination of exchange programs, and administrative activities.

REFERENCES

References available upon request.